

Time: 9.00-14.00. Limits for the credits 3, 4, 5 are 18, 25 and 32 points, respectively, including bonus points from hand-in assignments. The solutions should be well motivated.

Permitted aids: Pocket calculator. Dictionary. Formelsamling för inferensteori 1.

1. We have a random sample 1,3,4,3,4 from a random variable X with probability function

$$P(X = k) = \begin{cases} 1 - 10\theta, & k = 0, \\ \theta k, & k = 1, 2, 3, 4, \\ 0, & \text{otherwise,} \end{cases}$$

where $0 \leq \theta \leq 0.1$. Estimate θ using

- (a) the method of moments, (1p)
 - (b) the least squares method, (2p)
 - (c) maximum likelihood. (2p)
2. We have a random sample x_1, x_2 from a random variable X with expectation μ and variance σ^2 , and another random sample $y_1, y_2, \dots, y_5$ from a random variable Y with expectation 5μ and variance $5\sigma^2$. We may assume that X and Y are independent. The sample means are denoted $\bar{x}$ and $\bar{y}$.

The following estimates of μ are proposed:

$$\mu_1^* = \frac{\bar{x} + \bar{y}}{6}, \quad \mu_2^* = \frac{5\bar{x} + \bar{y}}{10}.$$

- (a) Show that μ_1^* and μ_2^* are both unbiased. (1p)
- (b) Which one of μ_1^* and μ_2^* is most efficient? (2p)
- (c) Is there any other unbiased estimate on the form $a\bar{x} + b\bar{y}$ which is more efficient than both μ_1^* and μ_2^* ? (2p)

3. Barry arranges a lottery with a large number of ballots (lotter). He claims that on average, one ballot out of five is a win.

Penny buys 25 ballots, but no one of them is a win.

- (a) Is Barry telling the truth, or is he cheating? Try to find this out by performing a hypothesis test. (1p)
 - (b) What is the critical region of the test in (a) if it is performed on the 5% level? (2p)
 - (c) What is the power of the 5% level test if in fact, on average only one ballot out of 50 is a win? (2p)
4. Helga goes by bicycle to work. She has two routes, A and B. She wants to know if the routes are equally fast. Everyday, she flips a coin to decide which route to take. After ending the experiment, she got the data given in the table below. The times are given in minutes and parts of minutes (hence not seconds).

Route A	10.27	10.72	10.36	10.01	10.25
Route B	9.86	9.99	10.02	10.22	9.52

Are the routes equally fast? Try to answer this question by performing a suitable statistical test. Make sure to specify all your assumptions. (5p)

5. A company produces a kind of electrical units. The number of defect units for one day is supposed to follow the Poisson distribution with parameter μ .

One day, the company produced 45 defective units.

- (a) Calculate a 99% confidence interval for μ . (2p)
 - (b) Test $H_0: \mu = 30$ vs $H_1: \mu \neq 30$ at the level 1% by using the test variable method. (2p)
 - (c) Compare the results in (a) and (b). Do they lead to the same conclusion? Explain. (1p)
6. Before the US president election, opinion polls were held in different states. For the state of Arizona, the poll of Survey/Minkey/Axios, Oct 20- Nov 2, gave Donald Trump 46% of the sympathies. The number of asked voters was 4278. At the same time, a corresponding poll in New Jersey gave Donald Trump 38% of the sympathies, where the number of asked voters were 3870.
- (a) Calculate a 95% confidence interval for the difference of proportions of Trump voters for the two states. (4p)
 - (b) Were the proportions of Trump voters different in the two states? (1p)

7. At the North Pole, where Santa Claus (jultomten) lives, his pixies (tomtenissar) have produced 10 000 baby dolls. These dolls are supposed to be identical, but nevertheless, their weights vary a little bit. The weights of the 10 000 baby dolls can be seen as a random sample with observed mean $\bar{x} = 237.00$ g and observed standard deviation $s = 1.20$ g.
- (a) Calculate a 95% confidence interval for the weight of a baby doll of the type produced by the pixies. (1p)
 - (b) Santa gets angry with the pixies if the proportion of dolls with smaller weight than 235 g is greater than 10%. Calculate a suitable confidence interval to try to conclude if Santa will get angry this time. You may assume that the weight of a baby doll is normally distributed. (4p)

8. The production of wind energy in Sweden (in GWh) over the years 1993-2015 is plotted in figure 1. As is seen from this figure, the production increases exponentially over time. In figure 2, instead the (natural) log of the production is plotted.

Let Y_t be the log of the wind energy production in Sweden for year t , and consider the regression model

$$Y_t = \alpha + \beta t + \varepsilon_t,$$

where $t = 1993, 1994, \dots, 2015$. The ε_t are assumed to be independent $N(0, \sigma^2)$.

This model was run for the data, and the following R output was produced:

Call:

```
lm(formula = y ~ t)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.42137	-0.16032	0.05747	0.17785	0.34174

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-476.30335	13.14537	-36.23	<2e-16 ***
t	0.24109	0.00656	36.75	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2087 on 21 degrees of freedom

Multiple R-squared: 0.9847, Adjusted R-squared: 0.984

F-statistic: 1351 on 1 and 21 DF, p-value: < 2.2e-16

- Which are the estimates of the α and β parameters, and what is the coefficient of determination here? (2p)
- Is there evidence that the wind energy production in Sweden changes over time? Motivate your answer. (1p)
- In figures 3-4, a histogram and a qq plot of the model residuals are given. Figure 5 plots the residuals (on the y axis) vs the fitted values (on the x axis). Based on these plots, do you think that the model is a good description of the data, and in that case, why (or why not)? (2p)

GOOD LUCK!

Appendix: figures

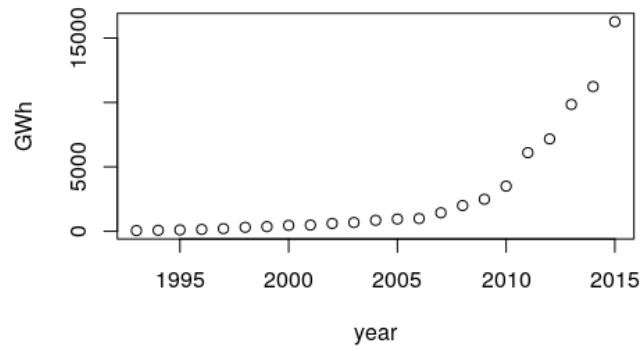


Figure 1: Swedish Wind energy production vs time.

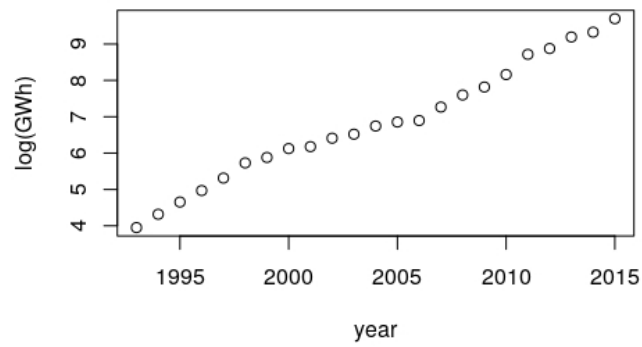


Figure 2: Log of Swedish Wind energy production vs time.

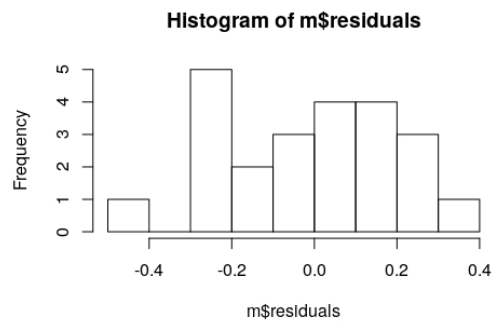


Figure 3: Histogram of residuals.

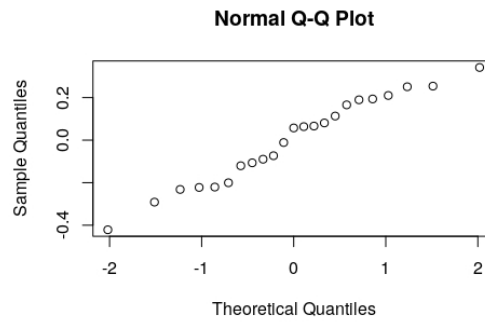


Figure 4: QQ plot of residuals.

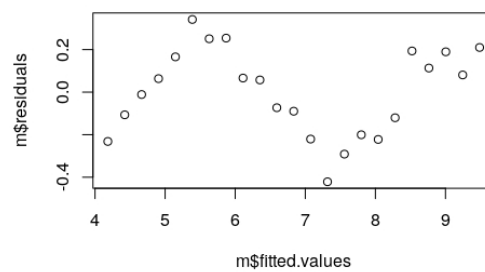


Figure 5: Residuals vs fitted values.